

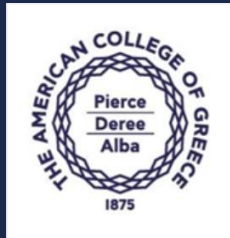
Customer Churn & Spending Prediction

in Telecommunications Using Machine Learning

ITC6103B1 · Applied Machine Learning · Winter Term 2026

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The Business Problem

IBM Telco · Acting as ML Strategy Consultants

Classification Target: Churn

Business question:

Will this customer leave?

Regression Target: Average Monthly Charges

Business Question:

What factors - demographic, connectivity, added services or contractual - drive customer spend?

Customer Retention

Classify at-risk customers before they leave, enabling proactive intervention campaigns.

Revenue Protection

Retain high-value customers and maintain market share with targeted offers.

Pricing Intelligence

Identify factors driving high/low billing and optimize contract features.

Spending Analytics

Detect customers with unusually high or low spend to inform strategy.

Dataset Overview

7,043

Total Records

21

Total Columns

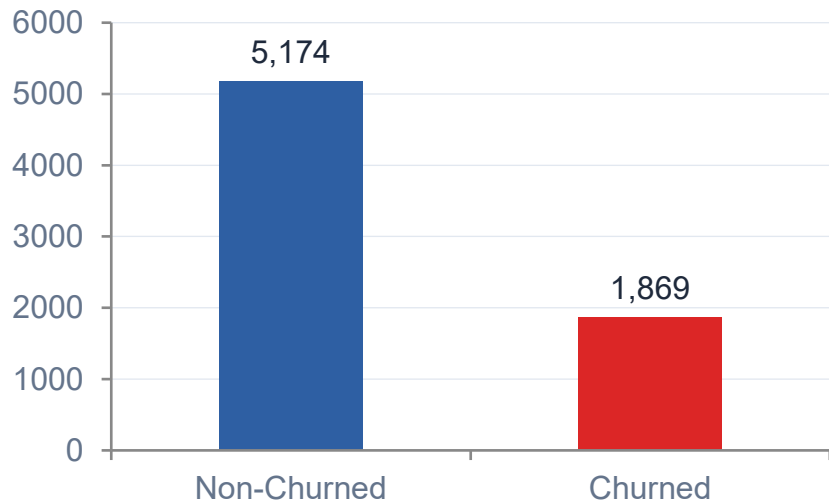
0

Duplicates

0

Missing Values

Class Distribution



Feature Breakdown

Numeric Features

3

Binary Categorical

7

Multiple Categorical

10

Data Preprocessing & EDA

01 High Cardinality

Dropped customerID (unique identifier). No useful predictive signal.

02 Binary Encoding

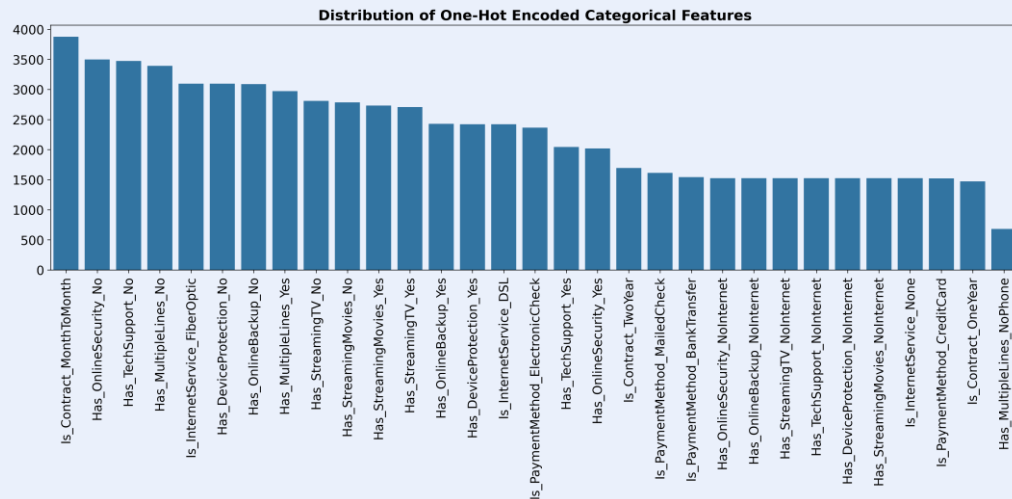
All binary features mapped to True/False using .map() (equivalent to 1/0 for ML)

03 OHE Encoding

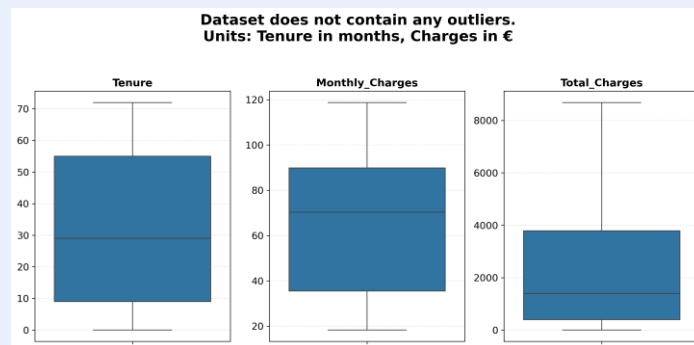
OHE instance applied, 2nd approach: get_dummies()

04 Total Charges Fix

11 empty values converted to valid nulls, then filled with 0 (€)



Outliers Checking, Feature Engineering and Feature Selection



Engineered Feature: $\text{Avg_Monthly_Charge} = \text{Total_Charges} \div \text{Tenure}$

Feature	vs Tenure	vs Monthly_Charge	vs Total_Charges
Avg_Monthly_Charge	0.249  Low	0.994  High (removed)	0.651 Moderate

Supervised Classification Analysis 70/30

Random Forest

Baseline

Train Acc: 0.9980

Test Acc: 0.7894

Strong overfitting (21pp gap). Fewer than half of churners identified.

Precision: 0.6355

F1 -score: 0.5501

Recall: 0.4848

ROC-AUC: 0.8216

Tuned (RandomizedSearchCV)

Train Acc: 0.8444

Test Acc: 0.7984

Overfitting resolved (4.6pp gap). Precision improved but recall still low.

Precision: 0.6675

F1 -score: 0.5581

Recall: 0.4795

ROC-AUC: 0.8416

SVM (RBF)

Baseline

Train Acc: 0.7673

Test Acc: 0.7402

`class_weight='balanced'` boosts recall to 0.78 — 60% higher than RF baseline.

Precision: 0.5069

F1 -score: 0.6088

Recall: 0.7807

ROC-AUC: 0.8199

Tuned (RandomizedSearchCV)

Train Acc: 0.7753

Test Acc: 0.7440

Recommended model. Optimal balance of recall, F1, and no overfitting.

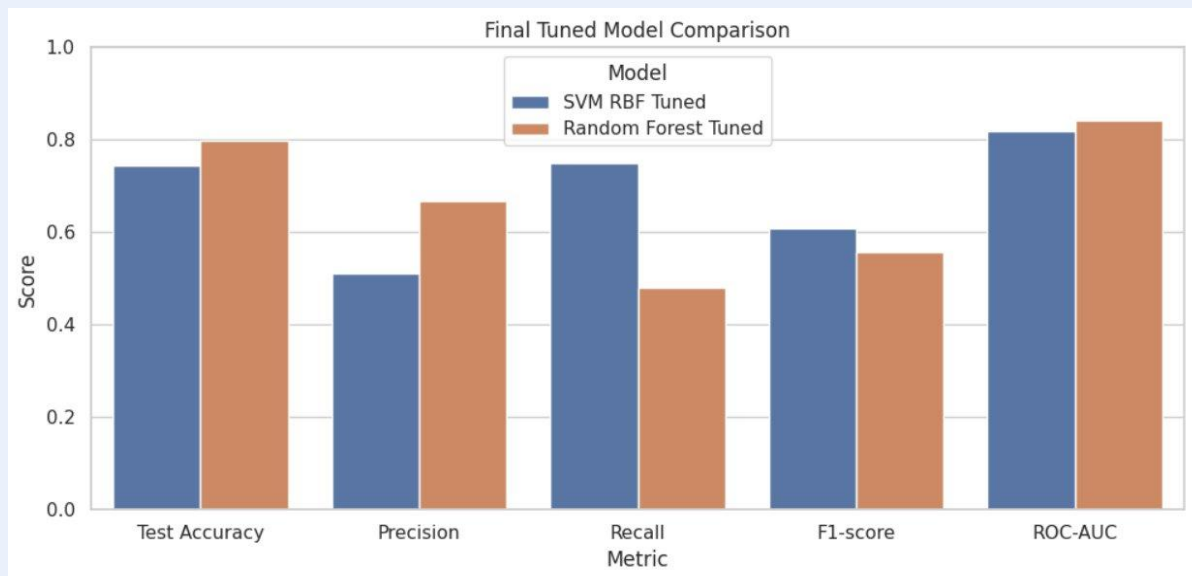
Precision: 0.5122

F1 -score: 0.6147

Recall: 0.7504

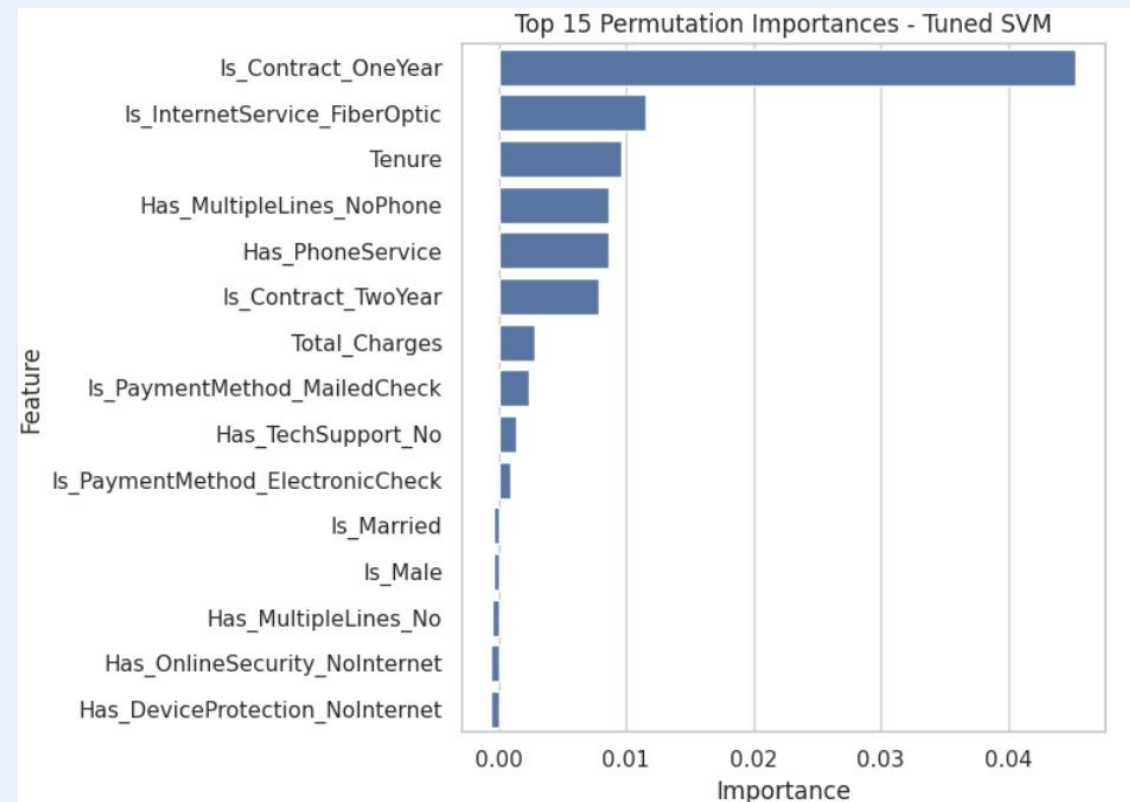
ROC-AUC: 0.8176

Classification Models — Comparison



Model	Train Acc	Test Acc	Precision	Recall	F1-Score	ROC-AUC
RF Baseline	0.9980	0.7894	0.6355	0.4848	0.5501	0.8216
RF Tuned	0.8444	0.7984	0.6675	0.4795	0.5581	0.8416
SVM Baseline	0.7673	0.7402	0.5069	0.7807	0.6147	0.8199
SVM Tuned	0.7753	0.7440	0.5122	0.7504	0.6088	0.8176

Why SVM Tuned is the Business Recommendation



75%

Churners Detected
(SVM Tuned)

48%

Churners Detected
(RF Tuned)

+152

Additional At-Risk Customers
Captured Per Cycle

Over a test set of 561 churned customers, the SVM correctly identifies ~421 vs ~269 for Random Forest.

Regression Analysis 70/30

Regression Analysis — Predicting Avg Monthly Charge

Key Setup Decisions

Target Variable	Avg_Monthly_Charge (Total_Charges / Tenure)
Dropped Features	Total_Charges (derived), Is_Churned (future info)
Dropped Rows	Tenure = 0 (11 new customers with no charges to predict)
Encoding after train/test split	ColumnTransformer with OHE for Categorical (drop='first') & StandardScaler for Numerical (Tenure)
Train/Test Split	70/30 split

Insight: All linear models achieve $R^2 = 0.991$. This indicates monthly charges are a direct additive sum of service subscriptions — exactly as expected for an itemised telecom bill. Dropping Tenure=0 rows improved results from $R^2=0.986$ to 0.991.

7 Baseline Models Compared

Model	R^2	MAE
Linear Regression	0.991	2.018
Lasso (a=0.01)	0.991	2.018
Ridge (a=0.1)	0.991	2.018
SVR Linear	0.991	2.020
XGBRegressor	0.990	2.122
Random Forest	0.988	2.337
SVR NonLinear	0.987	2.418
KNN	0.956	4.354

Regression Analysis — Feature Selection & Model Competition

Collinearity Fix

Removed **6 NoInternet**-suffix fields
(perfect correlation with
`Is_InternetService_None`)

Feature Engineering

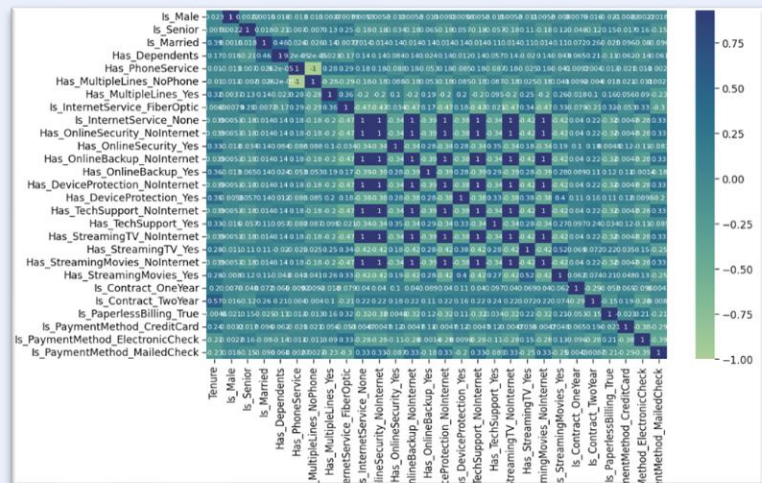
The **Household Type** as a
combination of `Is_Married` and
`Has_Dependents` : **Family, Couple,
Single Parent, Single Adult**

Remove Lasso zero-coefficient Features

Leaving us with **14 features** out of
the 28 features after OHE.

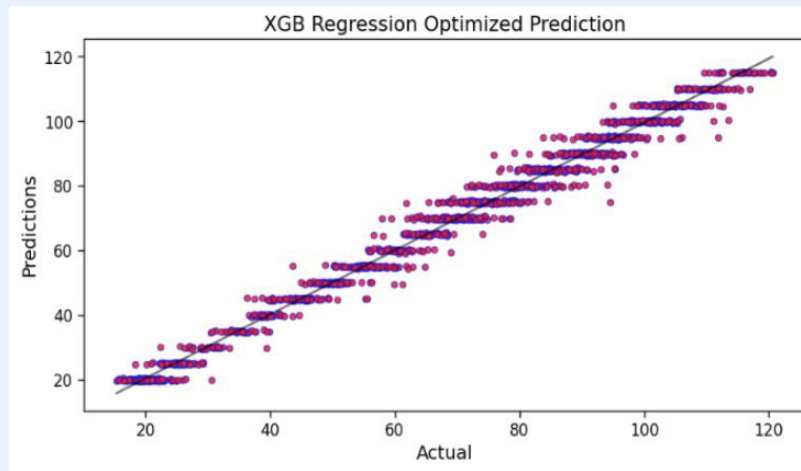
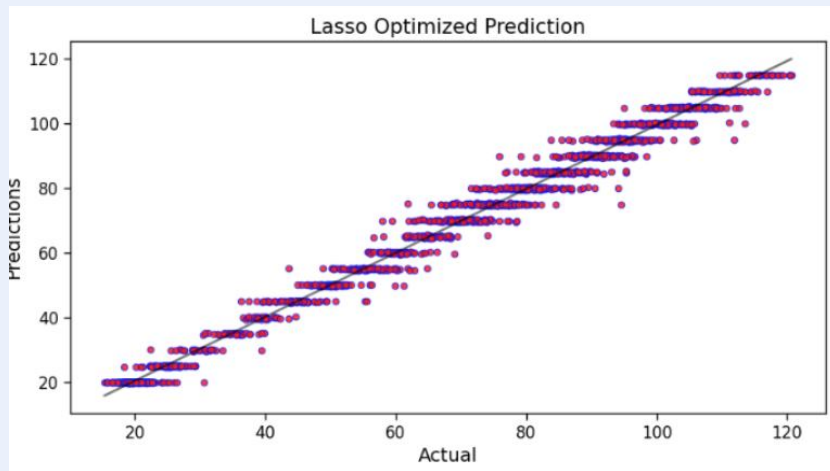
Randomized Grid Search for the XGBRegressor model

Created a parameter grid to identify
best parameters for the model.
Best Parameters : **max_depth = 1**



Model	R ²	MAE
Lasso 	0.991	2.018
XGBRegressor	0.990	2.120
XGBRegressor GridSearch	0.991	2.022

Regression Results — Lasso vs. XGBRegressor



General Insights

XGBRegressor

After grid search XGBoost model was "downgraded" to `max_depth = 1`

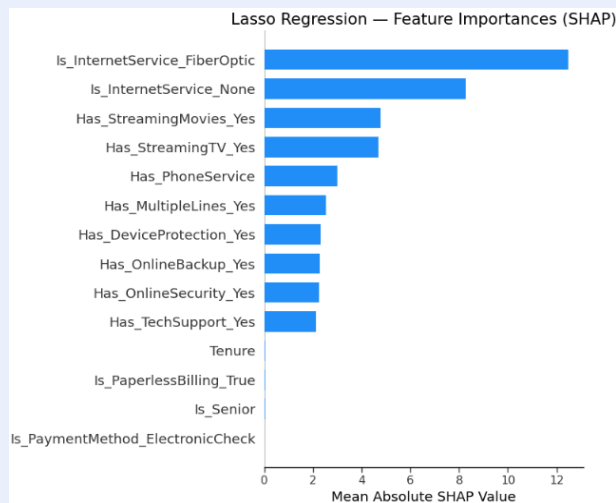
Avg Montly Spend

The billing relationship is linear. A complex model (XGBRegressor and Lasso arrived at the same result

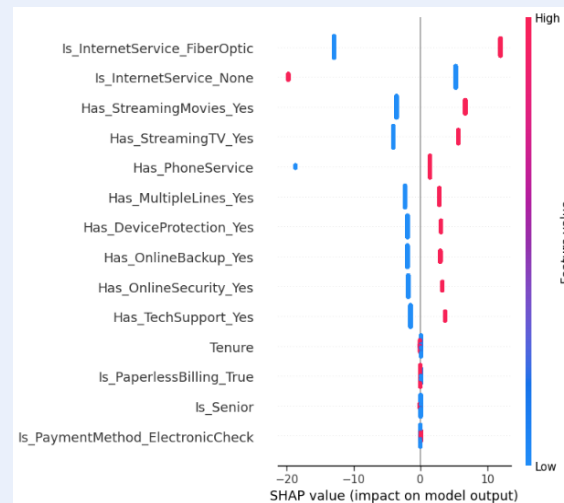
Scatter Plots - Predicted vs. Actual

Both Lasso and XGBRegressor capture the price drivers in a similar way

Regression Results — Model Explainability



LASSO SHAP LinearExplainer



LASSO SHAP Beeswarm

Top Drivers of Monthly Charges (Lasso / SHAP)

#1 Is_InternetService_FiberOptic + ~€12/mo

#2 Has_StreamingMovies / Has_StreamingTV +€5–8/mo

#3 Has_PhoneService / MultipleLines +€3–5/mo

#4 Is_InternetService_None –€20/mo

Additional Analysis

Additional Model Approach — 80/20 Split

Independent approach using XGBoost & Logistic Regression (classification) and XGBoost & Ridge (regression), with 80/20 train/test split.

Classification: XGBoost vs Logistic Regression

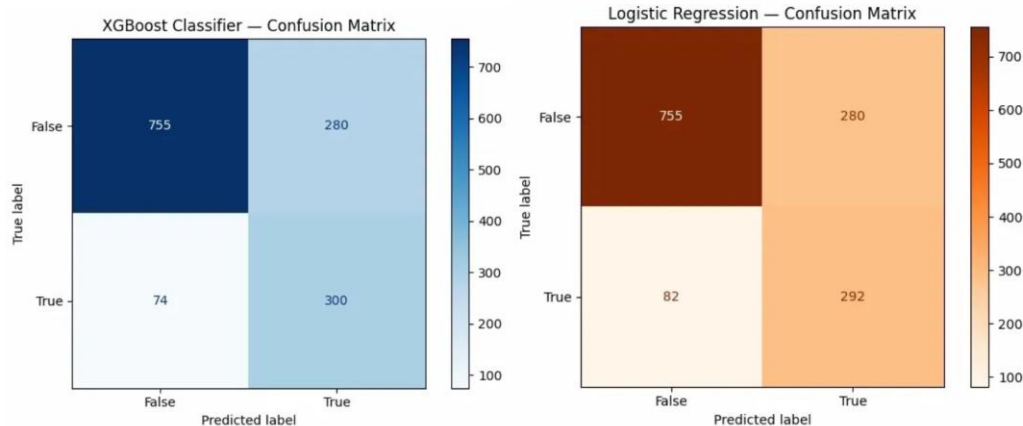


Figure 12 Confusion Matrix Comparison: XGBoost vs Logistic Regression

Regression: XGBoost vs Ridge

Metric	XGBoost	Ridge
R ² Test	0.9905	0.9871
MAE	2.09	2.04
Train R ²	0.9897	0.9877
Winner	R ²	MAE

SMOTE Experiment: Balancing the dataset with SMOTE reduced F1 from 0.6289 → 0.6197. The model overfit to synthetic rows rather than learning real patterns. Imbalance was handled inside the models instead: `scale_pos_weight` (XGBoost) and `class_weight` (Logistic Regression).

Takeaway: XGBoost Recall = 0.80 caught 300/374 churners. Logistic Regression was within 0.01 F1 — patterns are learnable even with simple models.

Additional Model Approach — 80/20 Split

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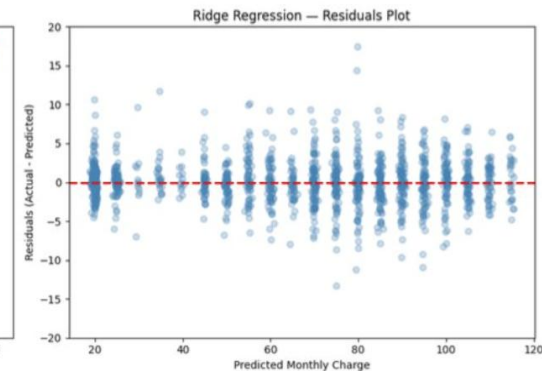
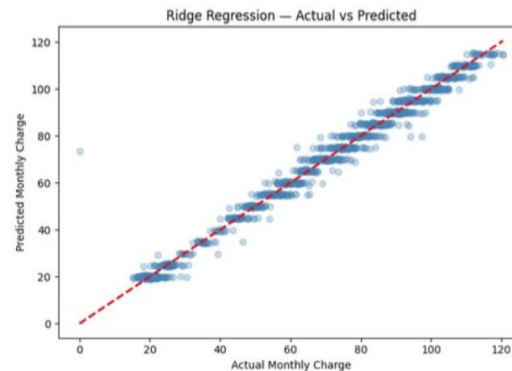
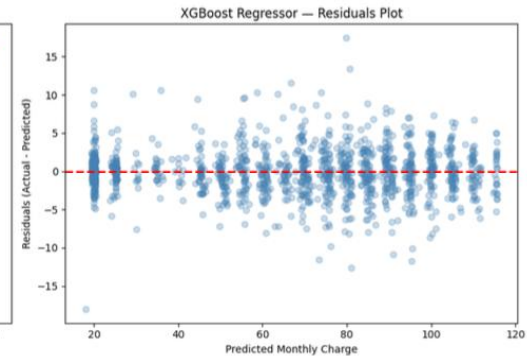
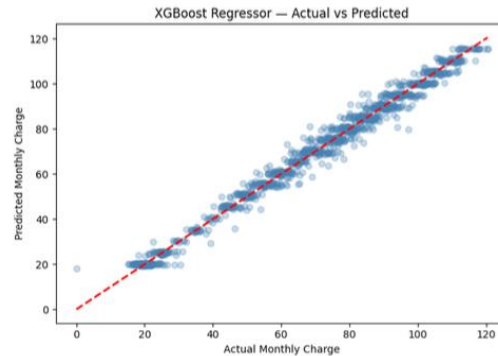


Figure 13 Regression Model Diagnostics: Actual vs Predicted and Residual Analysis

Explainable AI — SHAP Feature Importance

Churn Drivers (Classification)

#	Feature	XGB SHAP	LR SHAP
1	Tenure	0.62	0.72
2	Is_Contract_TwoYear	0.53	0.39
3	Avg_Monthly_Charge / FiberOptic	0.28	0.34

#1 Tenure

Longer tenure → less likely to churn

#2 Is_Contract_TwoYear

2-year contracts churn 15× less than MtM

#3 Avg_Monthly_Charge / FiberOptic

High bills on premium services = churn risk

Monthly Charge Drivers (Regression)

#	Feature	XGB SHAP	Ridge SHAP
1	Is_InternetService_FiberOptic	16.7	11.9
2	Has_OnlineSecurity_NoInternet	15.1	4.8
3	Has_StreamingMovies_No	4.0	4.6

#1 Is_InternetService_FiberOptic

Fiber Optic is the #1 driver of high bills

#2 Has_OnlineSecurity_NoInternet

No internet → significantly lower charges

#3 Has_StreamingMovies_No

Streaming add-ons drive higher bills

Conclusions & Business Impact

Classification

- ▶ Tuned SVM RBF is the recommended model
- ▶ Captures 75% of churners vs 48% (RF) — +152 customers/cycle
- ▶ XGBoost (80/20) achieved 80% recall independently
- ▶ Consistent across both approaches — findings are trustworthy
- ▶ Key churn signals: tenure, contract type, high bills

Regression

- ▶ All linear models achieve $R^2 = 0.991$, $MAE \approx 2.0$
- ▶ Monthly charges = direct sum of services (itemised billing)
- ▶ Fiber Optic is the #1 driver of high monthly charges
- ▶ Household_Type feature did not improve performance
- ▶ Simple Lasso nearly matches complex XGBoost

Thank You !

